Course III: Corporate Finance
Summer Term 2026
Unit 3 Mini-Cases

Please upload your solutions and indicate the Mini-Cases you have solved before 10:00 p.m. on April 06, 2026. There will be no deadline extensions!

To upload your solutions and make indications, please go to the assignment section on Canvas and choose "Unit 3 Indication and Submission of MCs". Then answer the true/false questions (Questions 1 to 6) and upload your solutions file (Question 7). Only complete submissions containing answers to the T/F questions and a solutions upload will be considered. Your last complete attempt counts.

You can either write your solutions using software or scan your handwritten solutions. Please note that you may only submit one solution file. If your solution consists of more than one file (e.g. 1 pdf and 1 Excel file), please upload a .zip file. Please make sure to hand in only one file of each file type (you might have to merge your scans to one pdf file). Uploaded files will be checked for plagiarism. The solutions in your file should be self-explanatory - not only for the instructor, but for other students as well. Indicate only Mini-Cases you feel confident to present.

General Assumptions

Unless stated otherwise, assume for all Mini-Cases that:

- All parties are risk-neutral.
- Management acts exclusively in the interest of existing shareholders.
- Perfect capital markets exist (no taxes and no transaction costs).
- The risk-free interest rate is constant for the duration of the case.

Mini Case 3.1

Petron Corporation is launching a new project. Depending on the project's success, one of the following free cash flows (FCF) will be realized one year from today:

State	Probability	FCF
1	10%	150
2	25%	120
3	35%	100
4	30%	70

Petron will not make any payouts to investors during the year. Assume a risk-free interest rate of 5% and that the project's risk is fully diversifiable.

Questions

a. What is the initial market value of Petron's equity without leverage?

b. Suppose Petron issues zero-coupon debt with a face value of 100 due next year.

   i. What is the value of the debt?

   ii. What is the value of equity with leverage?

   iii. What is the total value of the levered firm?

c. Now suppose that in the event of default, $\alpha = 25\%$ of the realized free cash flow is lost to bankruptcy costs.

   i. What is the value of the debt?

   ii. What is the value of equity with leverage?

   iii. Calculate the total value lost due to bankruptcy costs. Who bears these costs?

Mini Case 3.2

A company has existing assets with cash flows in one year distributed as follows:

State	**Probability**	**Cash Flow**
Good	40%	1,250
Bad	60%	450

The company has outstanding debt with a face value of 650 due in one year. The owners are considering a new project that requires an investment of 100 today and generates a certain cash flow of 150 in one year. Assume the risk-free rate is 5%.

Questions

a. Calculate the net present value (NPV) of the new project. Should the company invest in the project based on its NPV?

b. Are the equity holders willing to provide the company with the additional capital necessary for the investment in the new project?

c. Which specific agency cost of debt is illustrated by your answer to part (b)?

d. Suppose the outstanding debt face value is reduced so that the company only goes bankrupt if it does not invest in the project and the bad state occurs. What is the largest face value (F_{max}) at which the shareholders would be willing to invest?

e. Briefly explain why debtholders might be willing to accept a reduction in the face value of the company's outstanding debt.

f. Are the debtholders willing to agree to the face value of F_{max}?

g. What is the minimum face value (F_{min}) the debtholders would still accept?

Mini Case 3.3

A firm has outstanding debt with a face value of 40,000 due in one year. The company currently holds 5,000 in excess cash. It can either keep this cash safe (earning the risk-free rate of 5%) or invest the entire 5,000 today in a new project. Depending on the state of the economy in one year, the firm expects the following cash flows from its existing assets and the potential new project:

State	**Probability**	**Asset CF (CF_A)**	**Project CF (CF_P)**
Good	70%	40,000	6,750
Bad	30%	7,500	0

Questions

a. Calculate the current value of equity, debt, and the firm if the cash is kept safe.

b. Calculate the NPV of the new project.

c. Calculate the value of equity if the firm invests the cash in the project. Based on this result, does the management choose to carry out the project?

d. Calculate the value of debt and the total firm value if the project is implemented.

e. Explain the agency problem illustrated here. Who benefits from this investment?

Mini Case 3.4

A company is considering two mutually exclusive investment opportunities, both requiring an initial investment of 60 today. The potential realized cash flows in one year are:

	State	Probability	Cash Flow
Project 1 (Safe)	Good	80%	130
	Bad	20%	100
Project 2 (Risky)	Good	50%	200
	Bad	50%	20

Assume a risk-free interest rate of 0%.

Questions

a. Calculate the NPV of each project. Which project is optimal for the firm?

b. Calculate the value of equity for both projects if the firm is entirely equity-financed. Which project does the management choose?

Suppose the company finances the investment cost of 60 by taking out a loan. The lending bank knows the potential cash flows along with their probabilities, but it cannot observe which project the company actually chooses until after the loan is granted.

c. Suppose the bank assumes the safe project is chosen and charges a face value $F_{sp} = 60$. Which project does the management actually choose?

d. If the bank rationally anticipates management's choice in part (c), what face value (F_{be}) does it demand to break even?

e. Calculate the value of equity for both projects under the break-even face value (F_{be}) found in part (d).

f. Identify the agency problem illustrated here. How high is the expected agency cost of this debt financing?

g. Determine the maximum face value (F_{max}) that would leave the management indifferent between the safe and risky projects (i.e., the threshold that prevents risk-shifting). Assume the firm defaults only if the risky project fails. Briefly explain whether it is possible for the firm to raise financing for the safe project with standard debt under these terms.

Mini Case 3.5

Recall the setup and parameters from **Mini Case 3.4**. Arguing that it would solve the agency problem, the firm's management proposes a convertible debt contract for the loan amount of 60. The contract gives the lending bank the option to either receive the debt payment $F = 60$ or convert it into a share λ of the firm's total cash flows when the debt is due.

Questions

a. The management initially proposes a conversion rate of $\lambda_0 = 30\%$. Calculate the expected equity value under both projects. Which project does the management choose?

b. Does the bank accept the convertible debt contract?

c. Repeat the analysis from parts (a) and (b) assuming the conversion rate is increased to $\lambda_1 = 45\%$.

d. Determine the critical conversion rate $\lambda_0 < \lambda^* < \lambda_1$ for which the management is indifferent between the safe project and the risky one.

e. Would the bank accept the contract with the critical conversion rate λ^* found in part (d)? Briefly explain.

Mini Case 3.6

You have two job offers for a 2-year period. Assume a discount rate of 6% and that you only receive payments at the end of each year.

- **Firm A:**
  Pays 82,500 per year with certainty.
- **Firm B:**
  Pays 97,500 per year, but has a 60% chance of bankruptcy at the end of Year 1.

Should Firm B go bankrupt, you would receive a severance package (P) of 25,000 at the end of Year 2. You would also remain unemployed for 6 months (receiving 0) before starting a new job that pays an annual wage of 60,000. At the end of Year 2, you receive an amount exactly proportional to the time spent working at this new job.

Questions

a. Calculate the total present value of expected wages for Firm A.

b. Calculate the expected undiscounted wage in Year 2 associated with Firm B's offer, explicitly accounting for the probability of bankruptcy, the severance package, and the fractional year spent at the new job.

c. Calculate the total present value of expected wages should you choose Firm B.

d. Based on the calculated present values, which offer do you accept?

e. Determine for which bankruptcy probabilities you (weakly) prefer Firm A's contract.